



StudyStore

Online Stationery Shopping System

Prepared By :-

Kashyap Bharad - 92500584001

Dhaval Parmar - 92500584066

Chirag Pandya - 92500584070

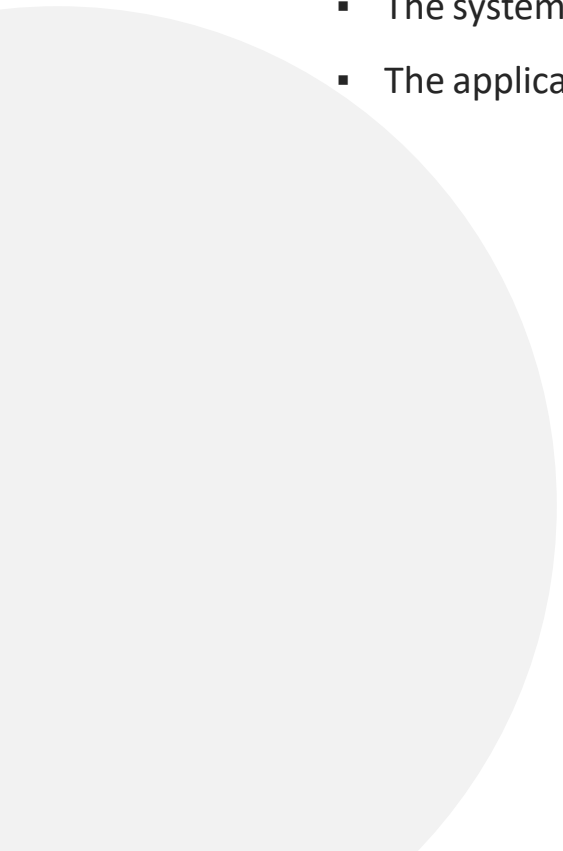
Internal Guide: Dr. Jaypalsinh Gohil

Introduction to Project

- StudyStore is a full-stack web application developed to provide an online platform for purchasing stationery products.
- Customers can browse products, search items, add products to their wishlist or shopping cart, and place orders through a simple web interface.
- Provides an administrator panel for managing products and customer orders.
- Built using the MERN Stack — React.js (Vite) for the frontend, Node.js and Express.js for the backend, and MongoDB Atlas for the database.
- JWT is used for user authentication, bcryptjs for password encryption, and Nodemailer for email-based password recovery.
- Aims to simplify the online purchase of stationery products through a dedicated, focused platform.

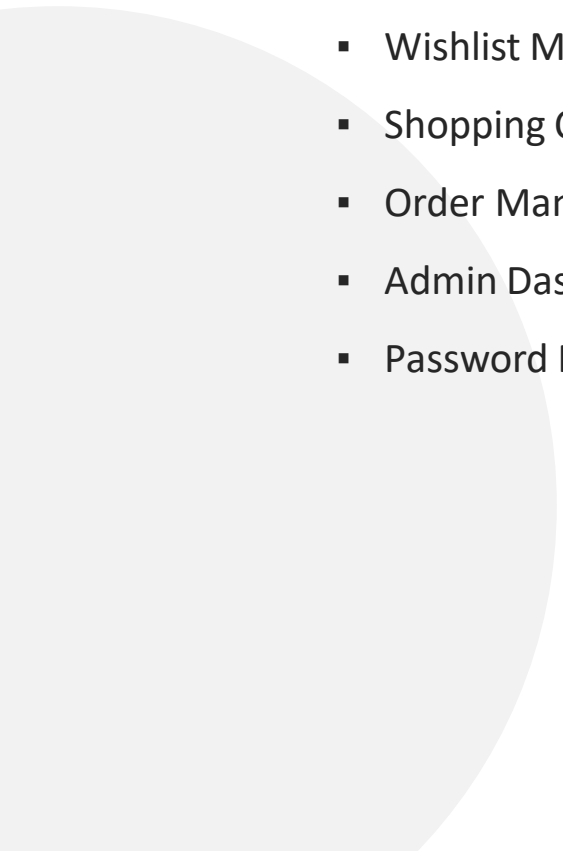


Preamble

- Many online shopping websites sell stationery products along with several other categories, making it difficult for users to quickly find what they need.
 - There is a need for a dedicated stationery shopping platform focused solely on stationery products.
 - StudyStore allows customers to browse, search, and place orders in a simple and organized environment.
 - The system also provides administrators with a centralized platform to manage products and customer orders.
 - The application demonstrates the practical implementation of full-stack web development using the MERN Stack.
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Module Description

- User Authentication Module
 - User Profile Module
 - Product Management Module
 - Search and Category Filter Module
 - Wishlist Module
 - Shopping Cart Module
 - Order Management Module
 - Admin Dashboard Module
 - Password Recovery Module
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Review of Literature

Purpose of Study

- Analyze existing online shopping and stationery platforms.
- Study their major features, product organization, and shopping workflows.
- Identify strengths and limitations of existing systems.
- Use the findings as a reference for developing StudyStore as a focused, dedicated platform.

Existing Systems Studied

- Amazon, Flipkart, Meesho (general e-commerce)
 - Scooboo, Odd Giraffe, Hindustan Stationery (dedicated stationery)
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Technical Description

Hardware Requirement

Component	Requirement
Processor	Intel Core i3 or Higher
RAM	4 GB Minimum (8 GB Recommended)
Storage	10 GB Free Disk Space
Display	1366 x 768 Resolution or Higher
Internet Connection	Required

Software Requirement

Software / Technology	Purpose
React.js + Vite	Frontend Development
Node.js + Express.js	Backend Development
MongoDB Atlas + Mongoose	Database Management
JWT + bcryptjs	Authentication & Password Encryption
Nodemailer	Email Service
Postman / Git & GitHub	API Testing & Version Control

System Design and Development

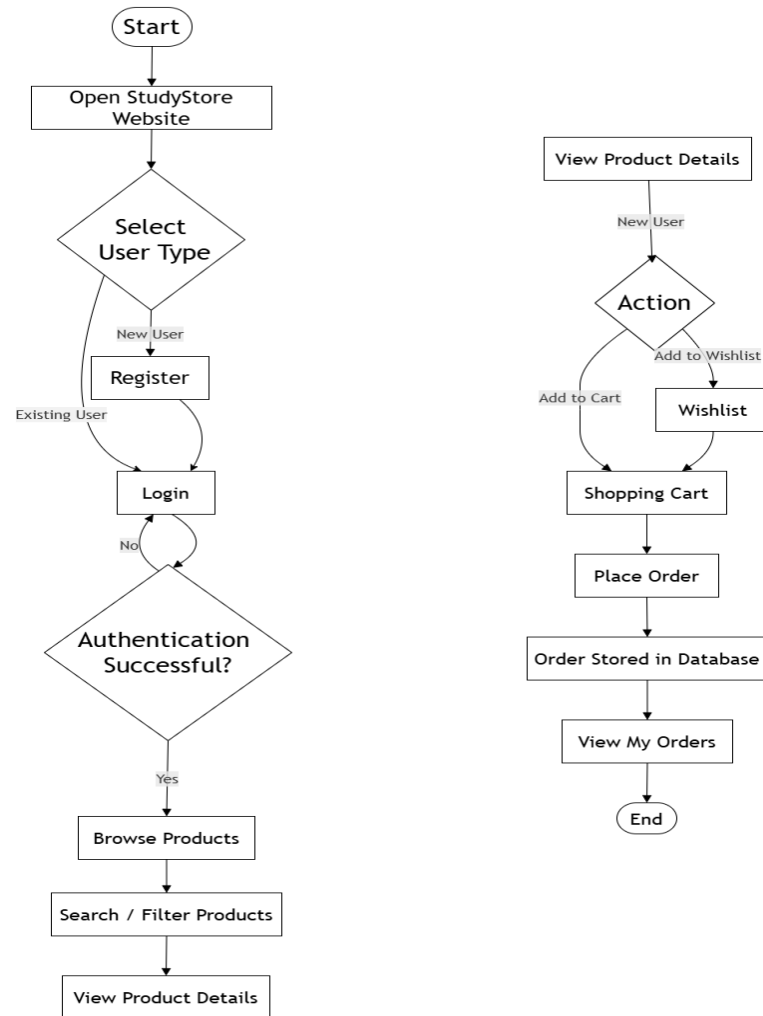


Figure 5.1: Customer Flowchart of StudyStore

System Design and Development

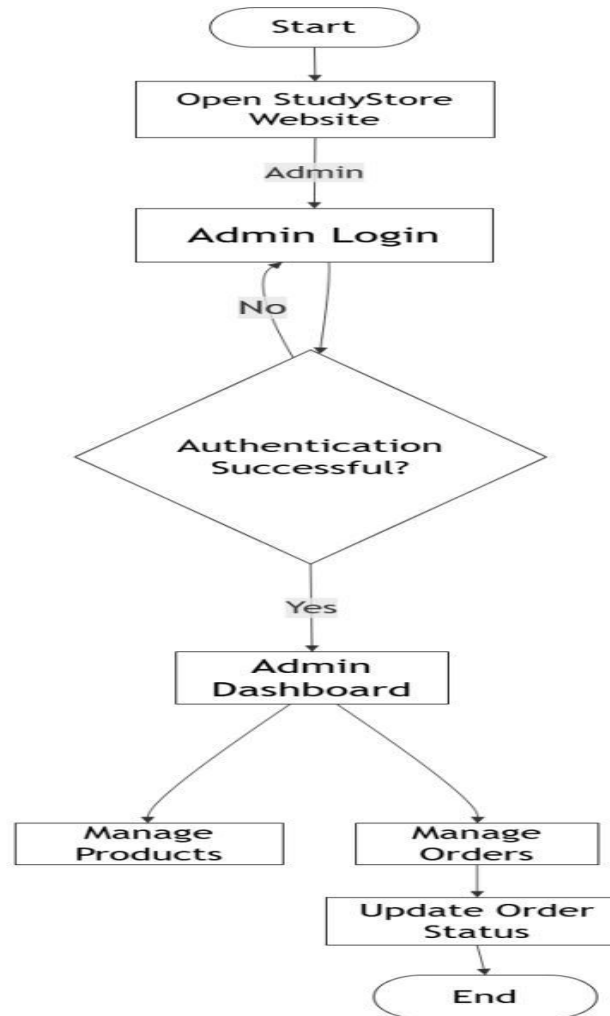


Figure 5.2: Administrator Flowchart of StudyStore

System Design and Development



Figure 5.3: Context Diagram of StudyStore

System Design and Development

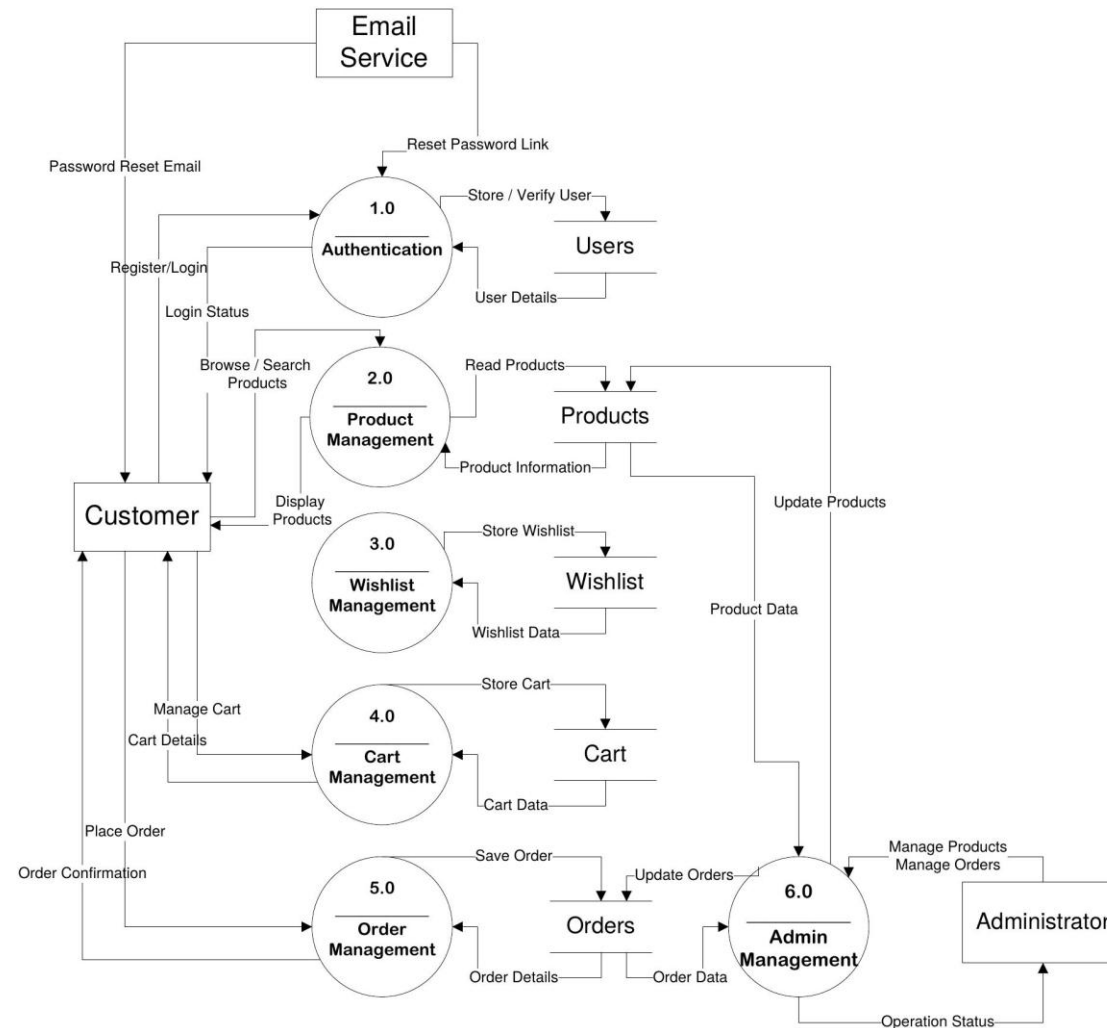


Figure 5.4: Level 1 Data Flow Diagram of StudyStore

System Design and Development

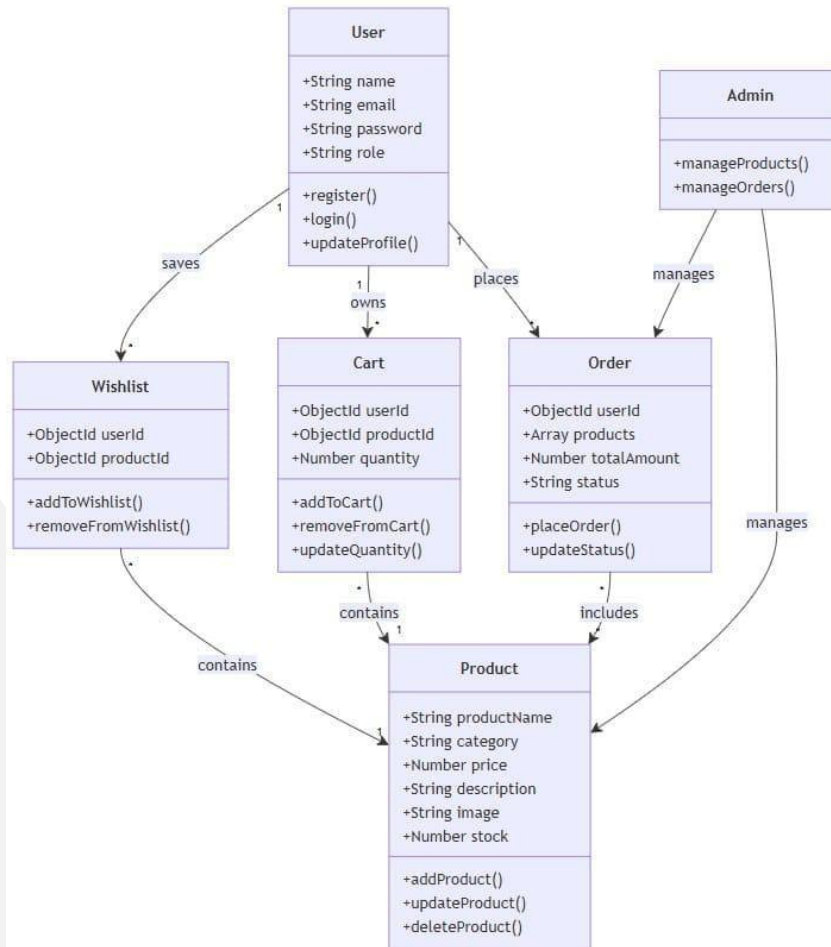


Figure 5.5: Class Diagram of StudyStore

System Design and Development

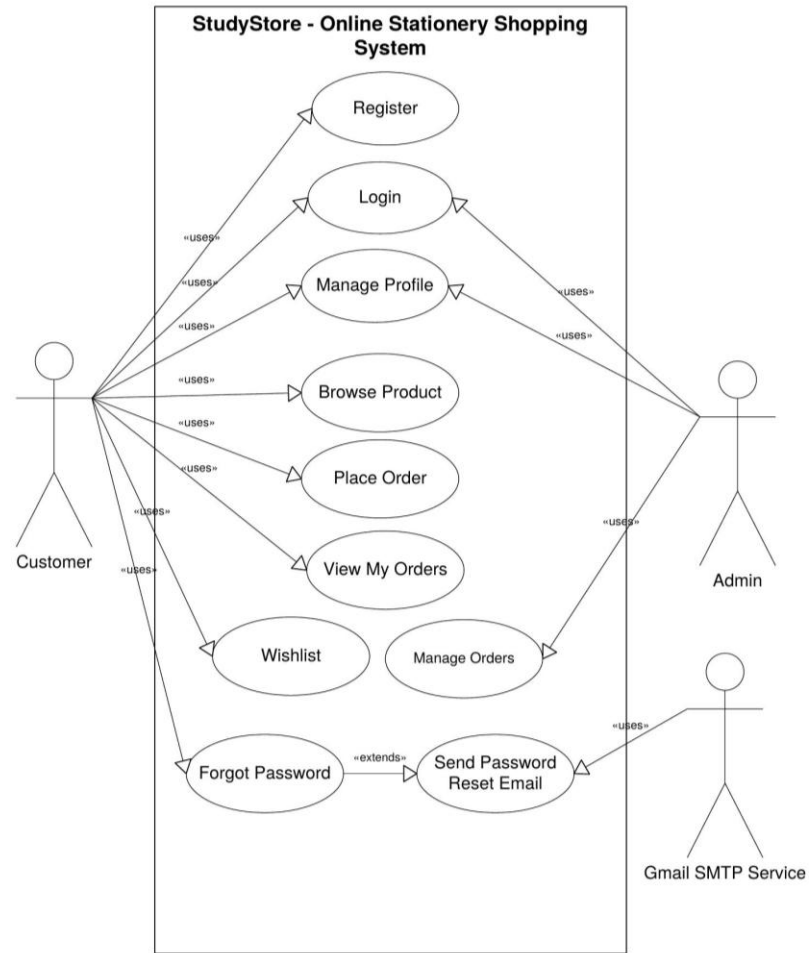


Figure 5.6: Use Case Diagram of StudyStore

System Design and Development

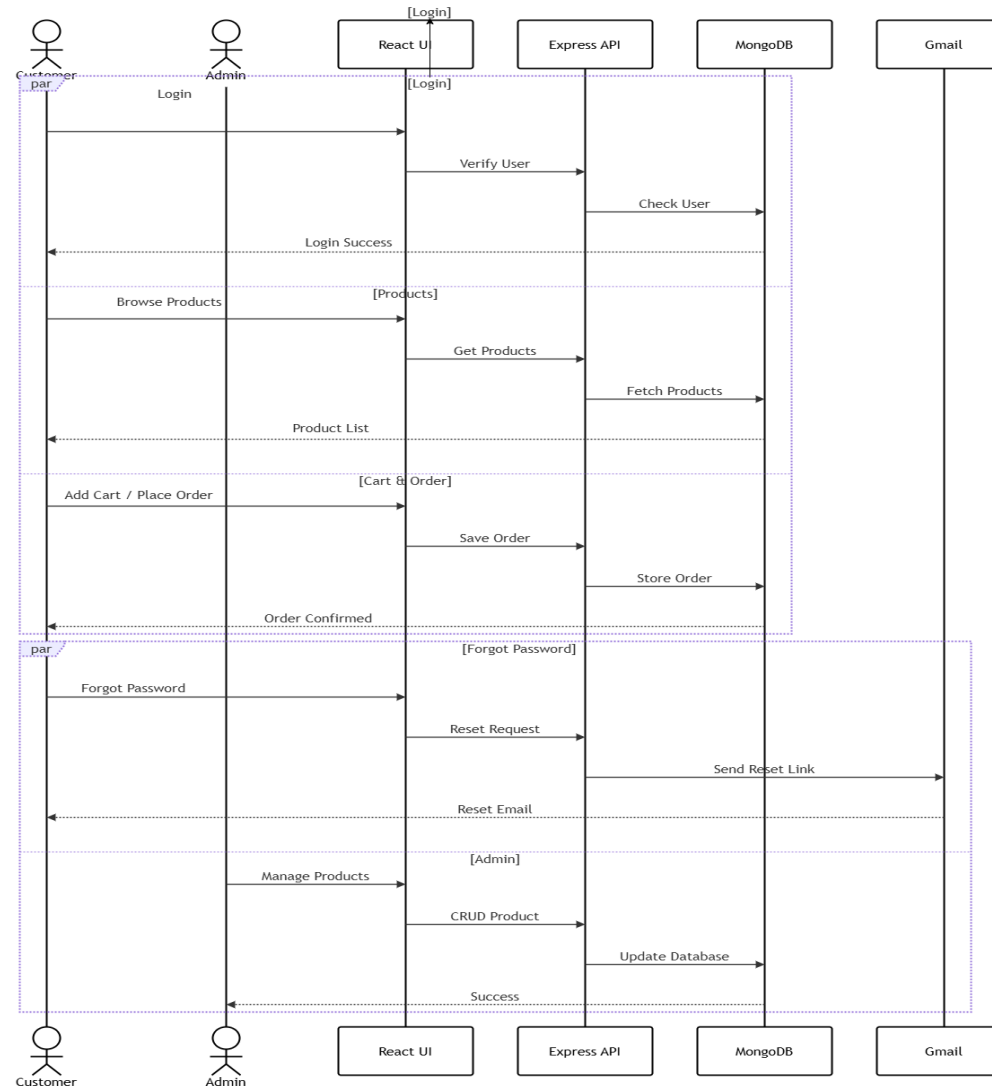


Figure 5.7: Sequence Diagram of StudyStore

System Design and Development

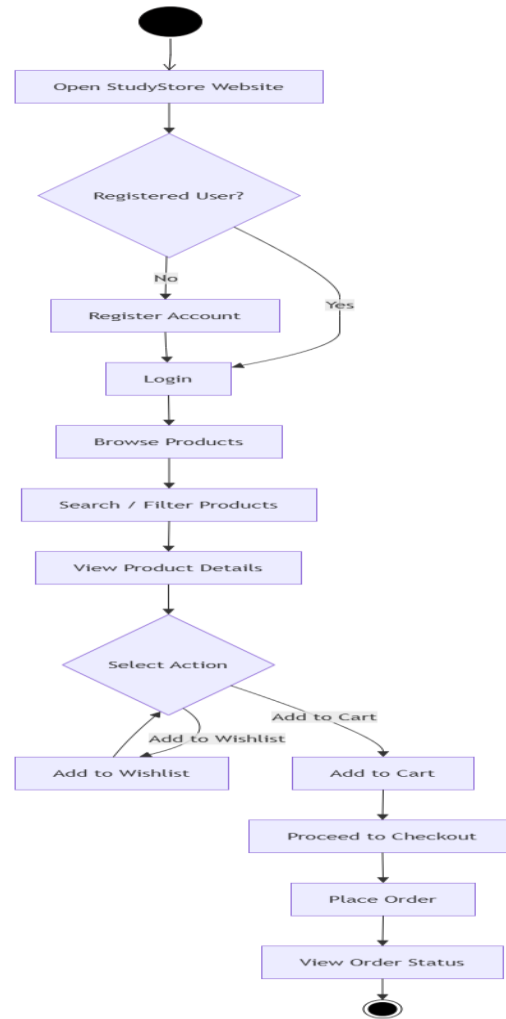


Figure 5.8: Customer Activity Diagram of StudyStore

System Design and Development

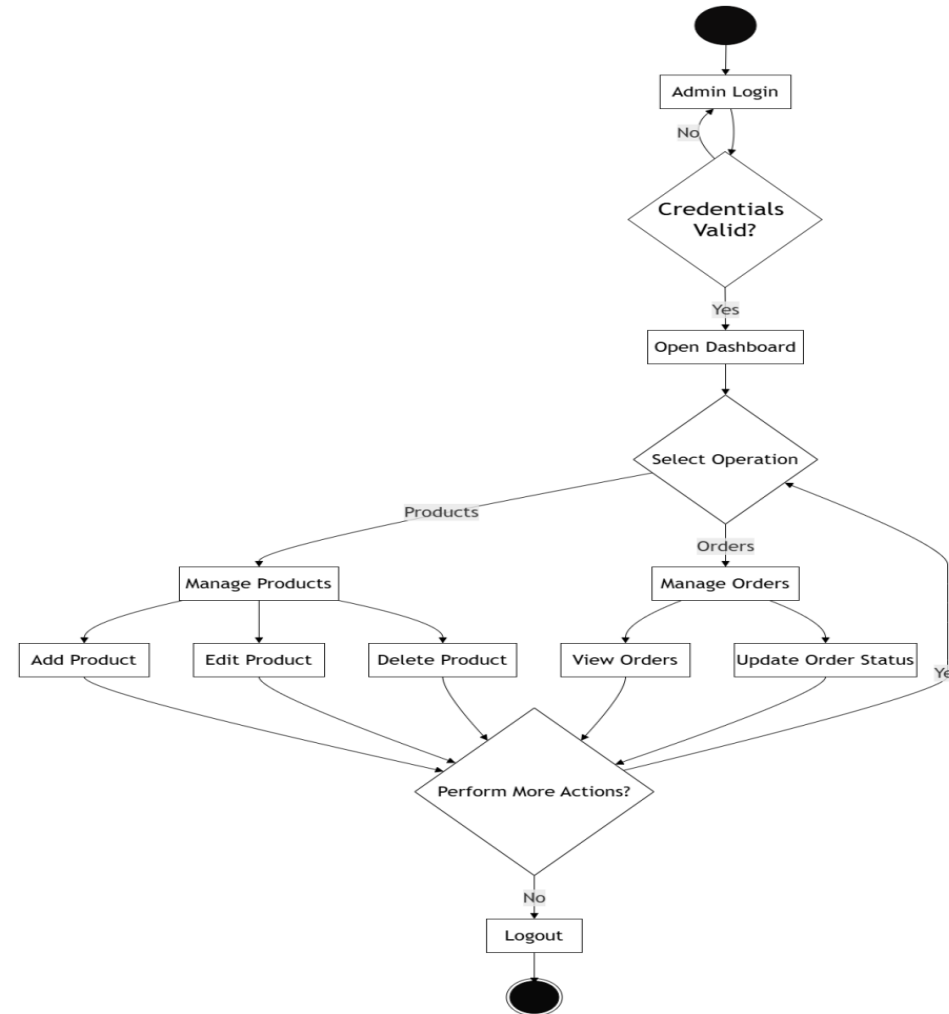


Figure 5.9: Administrator Activity Diagram of StudyStore

Database Design

Users Collection

Field Name	Data Type	Description
_id	ObjectId	Unique identifier of the user
name	String	Full name of the user
email	String	Unique email address
password	String	Encrypted password using bcryptjs
phone	String	Contact number
address	String	Delivery address
role	String	User role (User/Admin)
createdAt	Date	Account creation date

Database Design

Wishlist Collection

Field Name	Data Type	Description
_id	ObjectId	Unique wishlist identifier
userId	ObjectId	Reference to the user
productId	ObjectId	Reference to the product
createdAt	Date	Date when the product was added

Database Design

Products Collection

Field Name	Data Type	Description
_id	ObjectId	Unique product identifier
productName	String	Name of the product
description	String	Product description
category	String	Product category
price	Number	Product price
image	String	Product image path or URL
stock	Number	Available product quantity
createdAt	Date	Product creation date

Database Design

Cart Collection

Field Name	Data Type	Description
_id	ObjectId	Unique cart identifier
userId	ObjectId	Reference to the user
productId	ObjectId	Reference to the product
quantity	Number	Quantity selected by the user
totalPrice	Number	Total price of the selected quantity
createdAt	Date	Date when the product was added to the cart

Database Design

Orders Collection

Field Name	Data Type	Description
_id	ObjectId	Unique order identifier
userId	ObjectId	Reference to the user
products	Array	List of ordered products
totalAmount	Number	Total amount of the order
shippingAddress	String	Delivery address
paymentMethod	String	Payment method (Cash on Delivery)
orderStatus	String	Current order status
orderDate	Date	Date of order placement

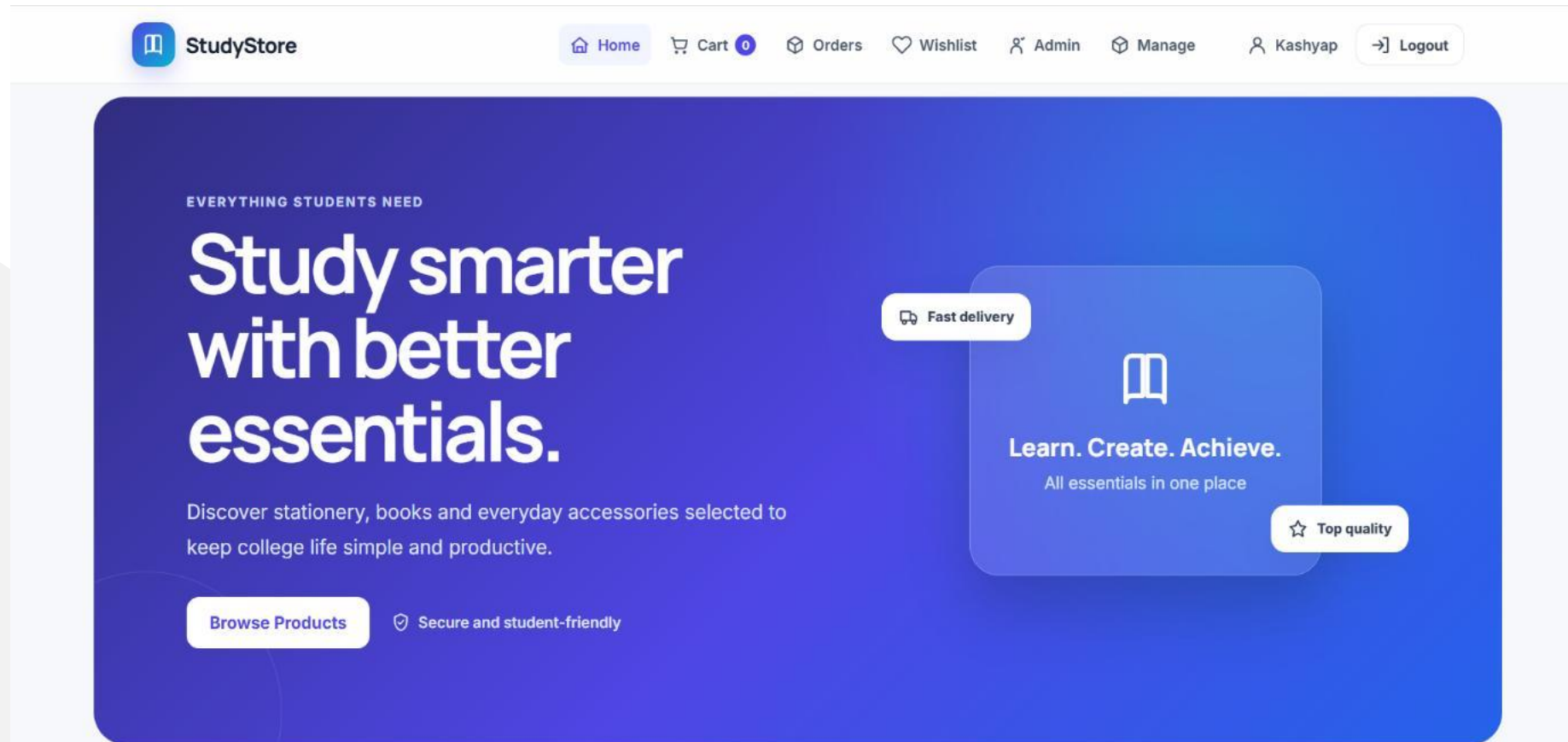
Database Design

Collection Relationships

Parent Collection	Child Collection	Relationship
Users	Wishlist	One user can save multiple wishlist items
Users	Cart	One user can have multiple cart items
Users	Orders	One user can place multiple orders
Products	Wishlist	One product can appear in multiple wishlists
Products	Cart	One product can appear in multiple shopping carts
Products	Orders	One product can be included in multiple orders

Screen Design

Home Screen



Screen Design

Registration Screen

 StudyStore

[Home](#) [Cart 0](#) [Orders](#) [Wishlist](#) [Login](#) [Register](#)

Create Account

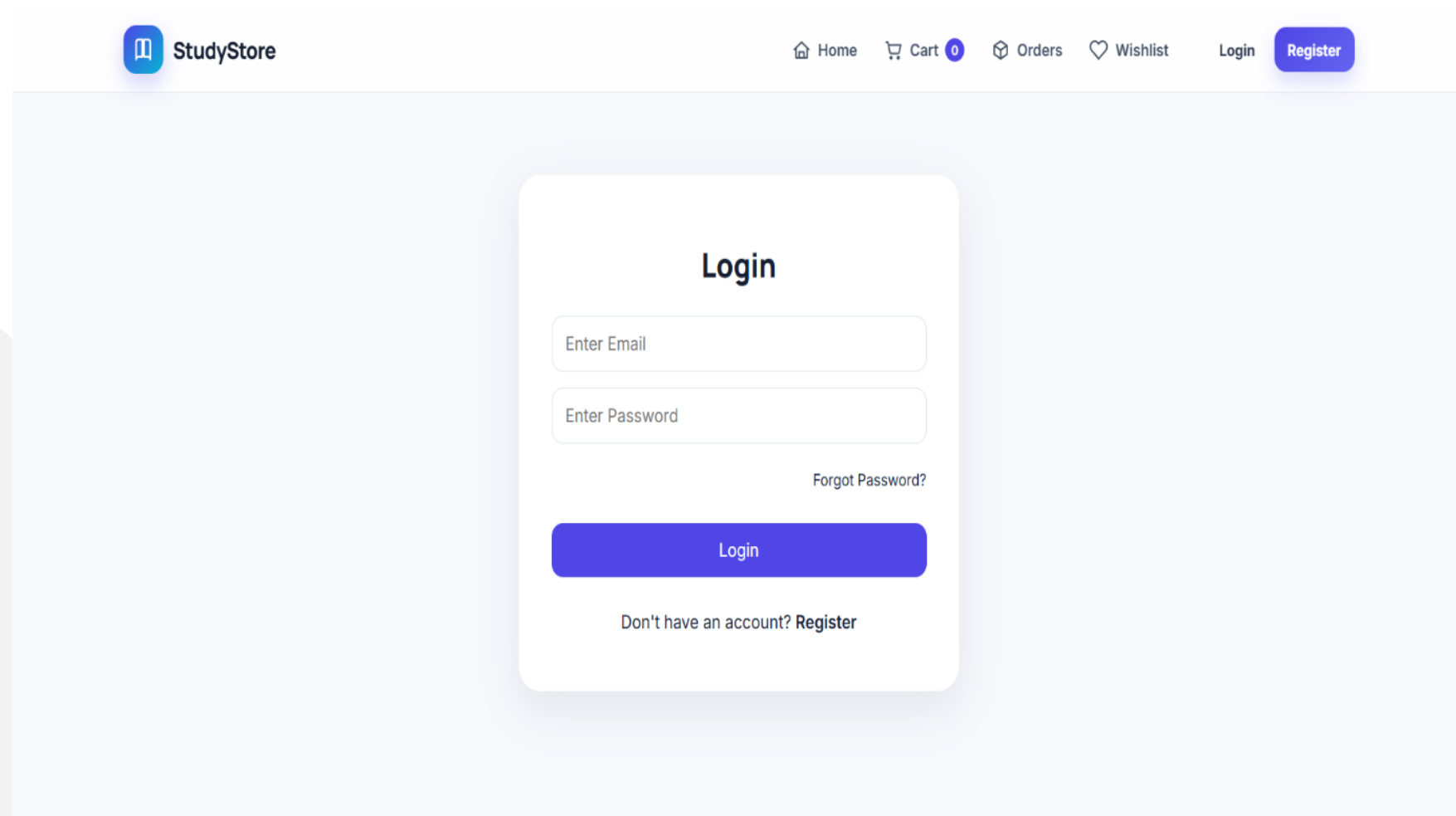
Register once and your delivery details will be ready at checkout.

[Send Email OTP](#)

[Activate Account](#)
Go to Settings

Screen Design

Login Screen



The image shows a web application login screen for 'StudyStore'. The header features the 'StudyStore' logo on the left and navigation links for Home, Cart (with a 0 item indicator), Orders, and Wishlist on the right. The 'Login' and 'Register' buttons are highlighted in blue. The main content area is a light blue rectangle containing a white login card. The card has a title 'Login', two input fields for 'Enter Email' and 'Enter Password', a 'Forgot Password?' link, a blue 'Login' button, and a link to 'Register' for users without an account.

StudyStore

[Home](#) [Cart 0](#) [Orders](#) [Wishlist](#) [Login](#) [Register](#)

Login

Enter Email

Enter Password

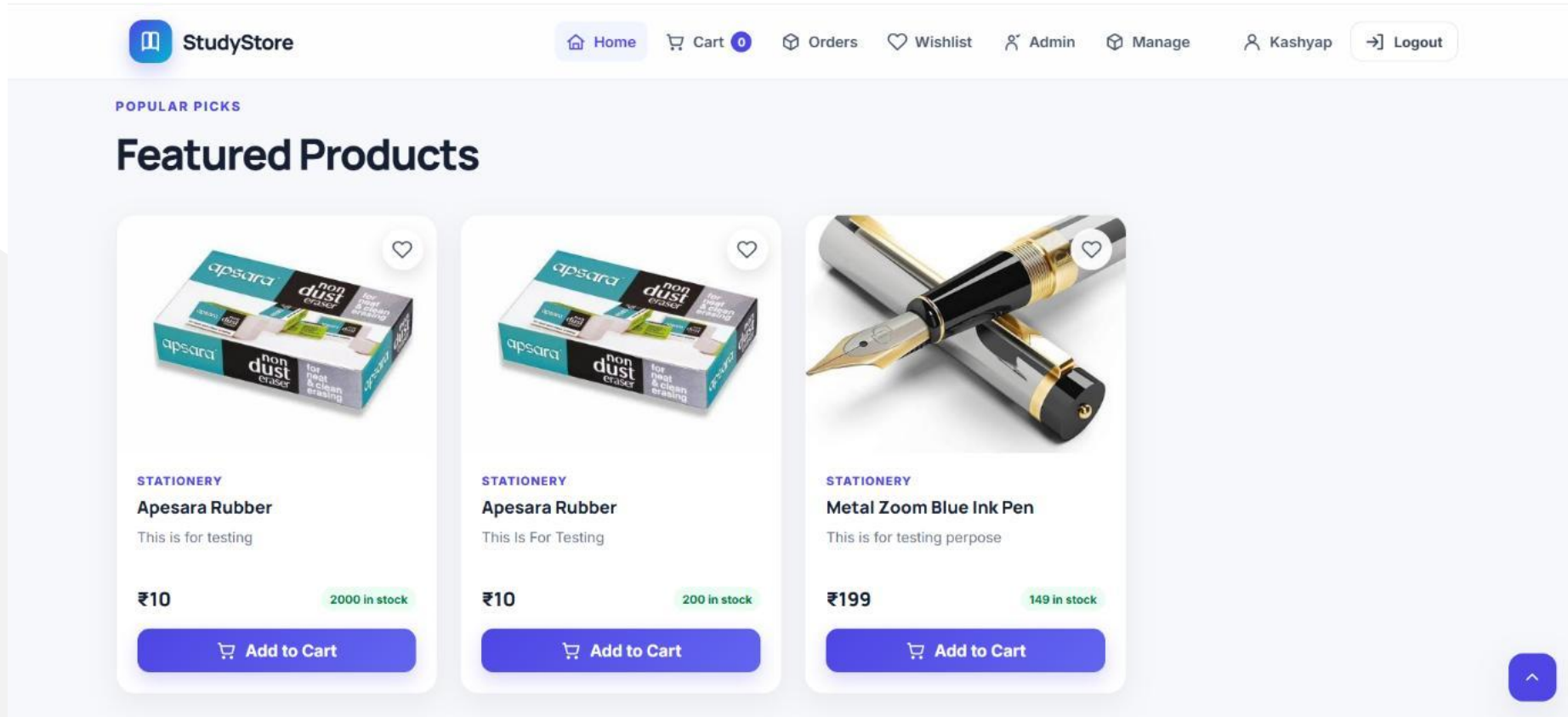
[Forgot Password?](#)

[Login](#)

Don't have an account? [Register](#)

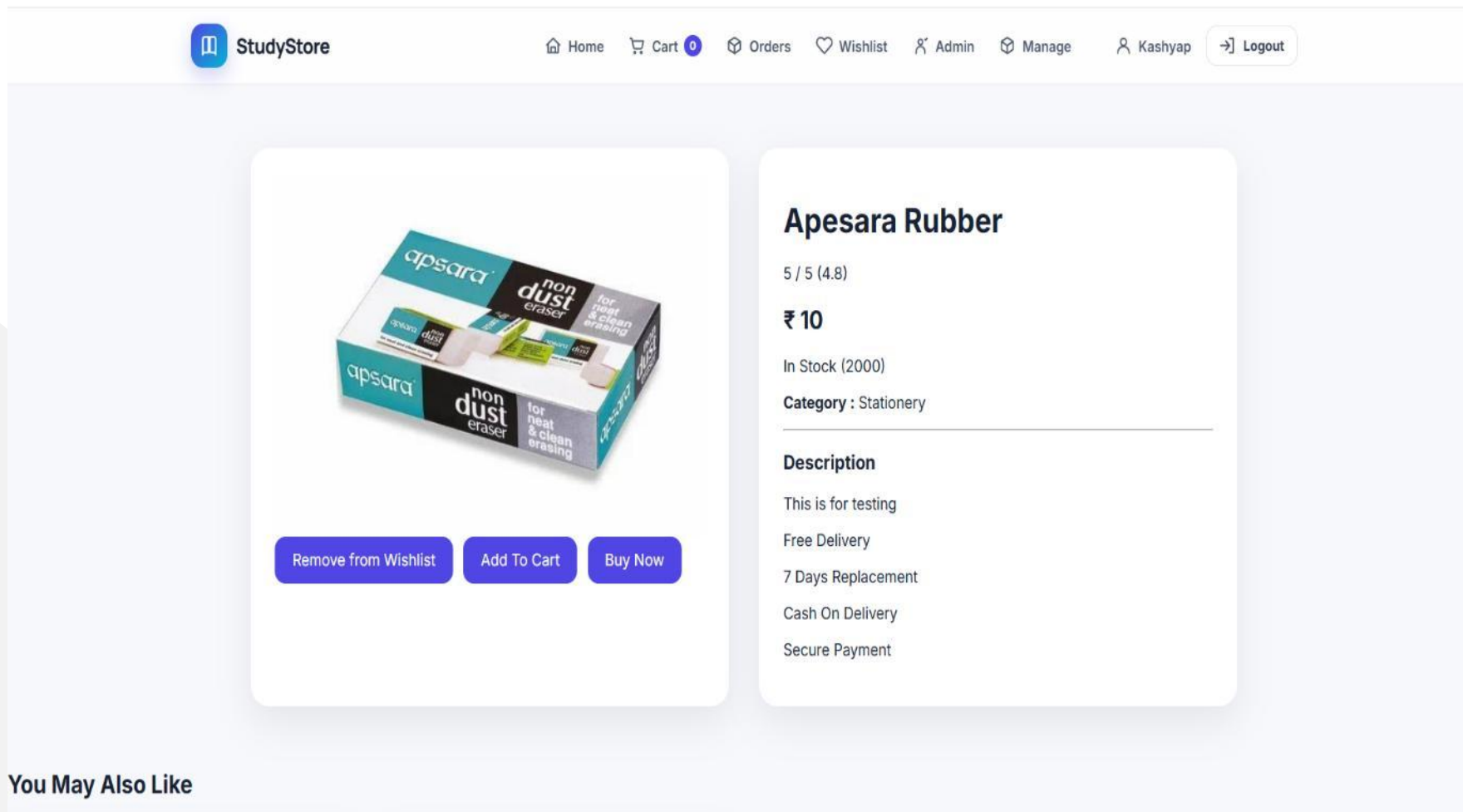
Screen Design

Product Listing Screen



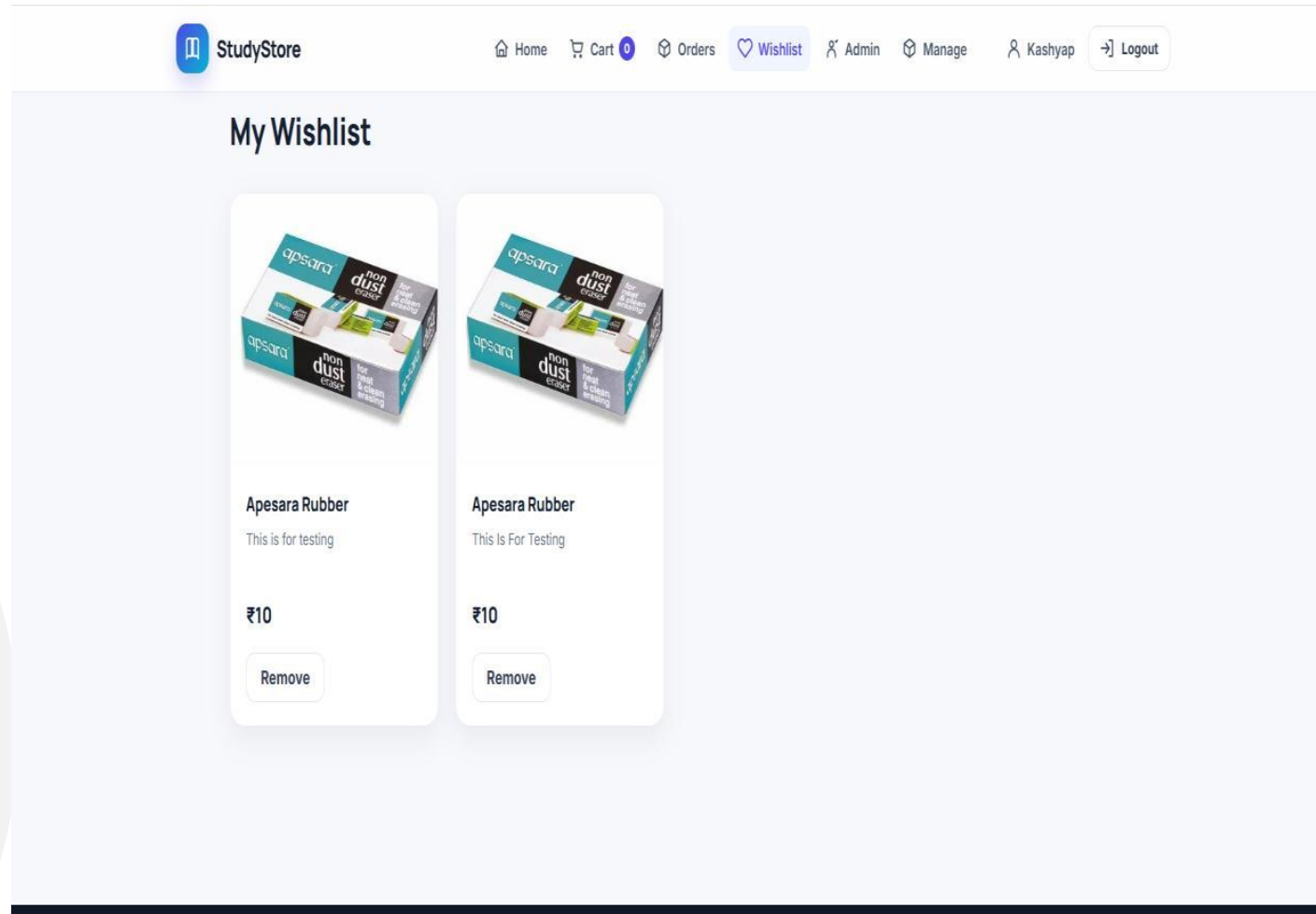
Screen Design

Product Details Screen



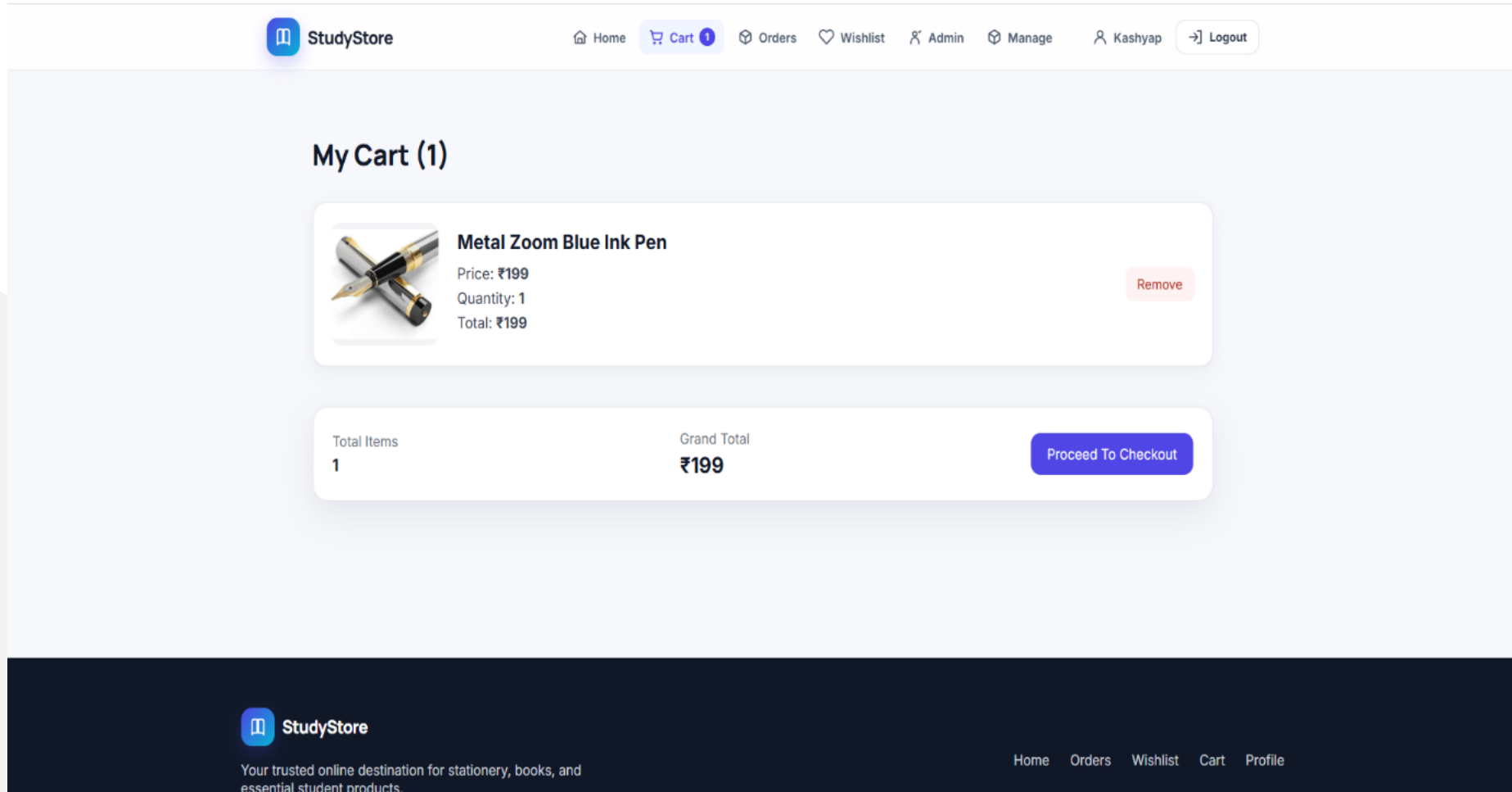
Screen Design

Wishlist Screen



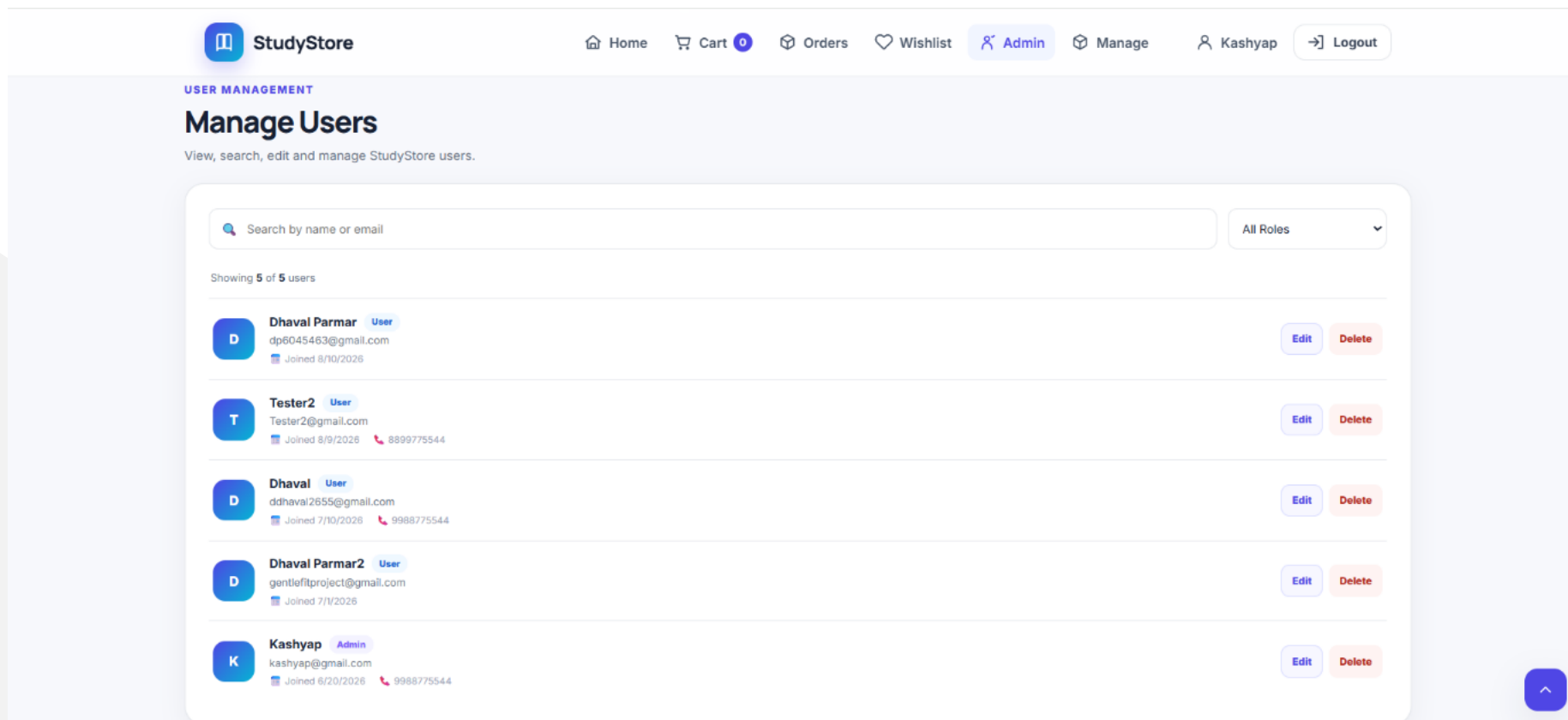
Screen Design

Shopping Cart Screen



Screen Design

User Management Screen



Screen Design

My Orders Screen

 StudyStore

[Home](#) [Cart 1](#) [Orders](#) [Wishlist](#) [Admin](#) [Logout](#)

My Orders

Order #5cc1b5 **Cancelled**

8/21/2026, 10:21:26 PM

Apesara Rubber × 1

₹10

Cash On Delivery · Pending

Rajkot, Gujarat

₹10

Order #5cc191 **Cancelled**

8/21/2026, 10:20:14 PM

Metal Zoom Blue Ink Pen × 3

₹597

Apesara Rubber × 3

₹30

Cash On Delivery · Pending

Rajkot, Gujarat

₹627

Order #3fa187 **Delivered**

8/14/2026, 10:02:04 PM

Apesara Rubber × 1

₹10

Cash On Delivery · Paid

Rajkot, Gujarat

₹10

Order #3fa17e **Cancelled**

8/14/2026, 9:59:07 PM

Metal Zoom Blue Ink Pen × 2

₹398

PayPal · Refunded

Rajkot, Gujarat

₹398

Activate Windows
Go to Settings to activate

Screen Design

Profile Screen

 StudyStore

[Home](#) [Cart 0](#) [Orders](#) [Wishlist](#) [Admin](#) [Manage](#) [Kashyap](#) [Logout](#)



Kashyap
kashyap@gmail.com

Edit Profile

 Save Profile

Screen Design

Forgot Password Screen

 StudyStore

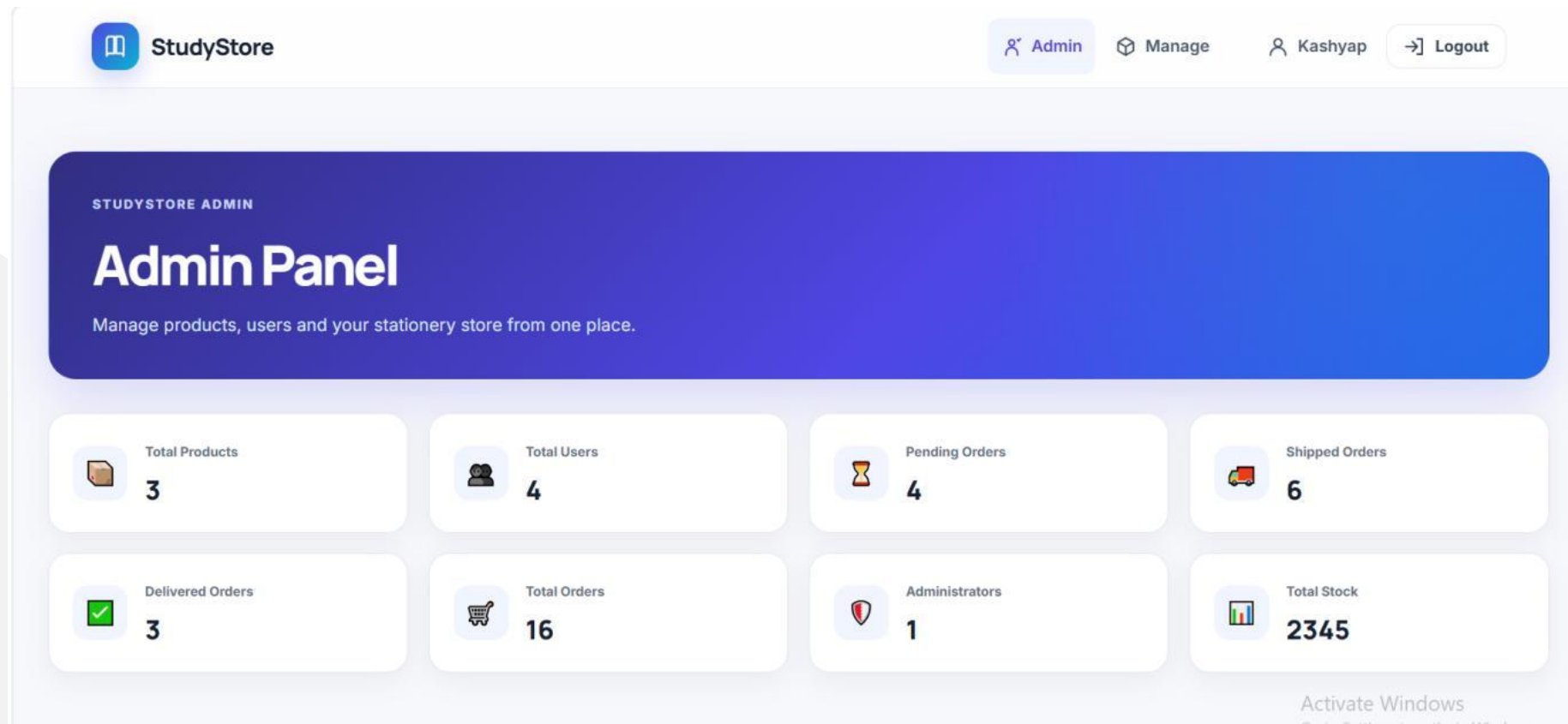
[Home](#) [Cart 0](#) [Orders](#) [Wishlist](#) [Login](#) [Register](#)

Forgot Password

Send Reset Link

Screen Design

Admin Dashboard Screen



Screen Design

Product Management Screen

 StudyStore

Admin

Manage

Kashyap

Logout

PRODUCT MANAGEMENT

Manage Products

Add new stationery products and manage your existing inventory.

Add New Product

Add a product to your store.

Product Name

Enter product name

Description

Enter product description

Price

₹ Price

Stock

Stock quantity

Category

Select category

Select Product Image

JPG, PNG or WEBP • Max 2 MB

+ Add Product

All Products

3 products currently available.



Apesara Rubber

STATIONERY

₹10 · Stock 1997

Delete



Apesara Rubber

STATIONERY

₹10 · Stock 199

Delete



Metal Zoom Blue Ink Pen

STATIONERY

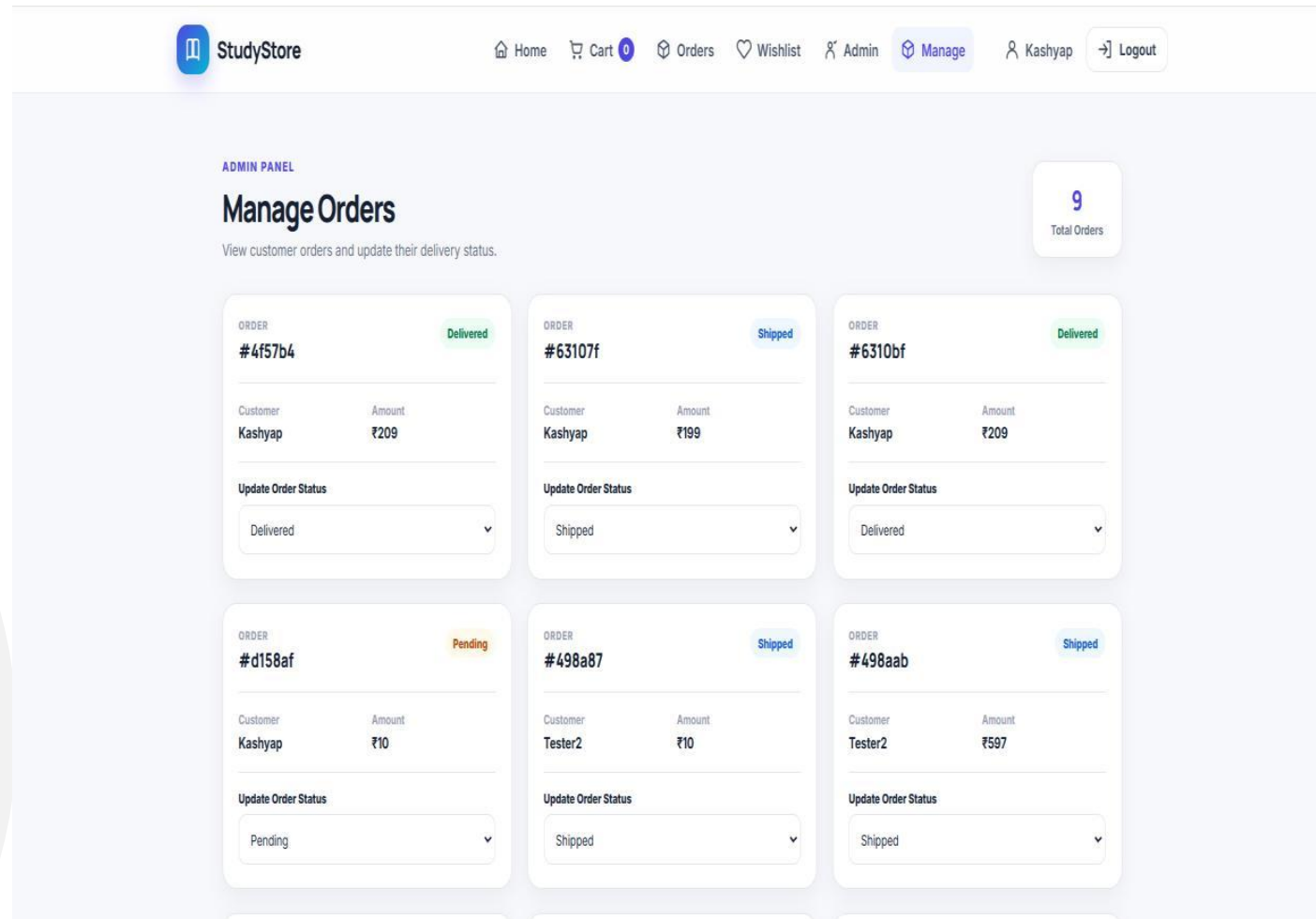
₹199 · Stock 149

Delete

Activate Windows
Go to Settings to activate Windows.

Screen Design

Order Management Screen



Screen Design

Coupon Management Screen

 StudyStore

Admin

Manage

Kashyap

Logout

Manage Coupons

Create discount coupons and manage existing promotional offers.

%

Create Coupon

Create a new discount coupon.

Coupon Code

WELCOME10

Discount Type

Percentage

Discount Value

10

Minimum Order

0

Maximum Discount

0 = unlimited

Expiry Date

dd-----yyyy --:-- --

☒ Coupon is active

Allow customers to use this coupon

+ Create Coupon

🏷️

All Coupons

1 coupons available.

%

WELCOME10

10% OFF

Min order ₹50 · No max discount

Expires: 9/15/2026, 12:00:00 AM

Active

Delete

Activate W
Go to Settings

Screen Design

Checkout Screen

 StudyStore

[Home](#) [Cart 1](#) [Orders](#) [Wishlist](#) [Admin](#) [Logout](#)

SECURE CHECKOUT

Complete your order

Enter your delivery details, choose payment and review your items.

1

Delivery Details

Your profile details are filled automatically. You can edit them here.

Full Name

Phone Number

Admin

9988776655

Complete Address

Ramapir chowk

City

State

Pincode

Rajkot

Gujarat

360007

Payment Method

Choose one payment option

Cash on Delivery

Pay when your order arrives

☐

UPI

UPI

Pay securely using Razorpay UPI

☒

Secure Payment

You will be redirected to Razorpay's secure UPI checkout after clicking Pay Now.

2

Order Summary

Review your items before payment.

 Apesara Rubber

₹10 × 1

₹10

Coupon Code

ENTER COUPON

Apply

Subtotal

₹10

Delivery

FREE

Total

₹10

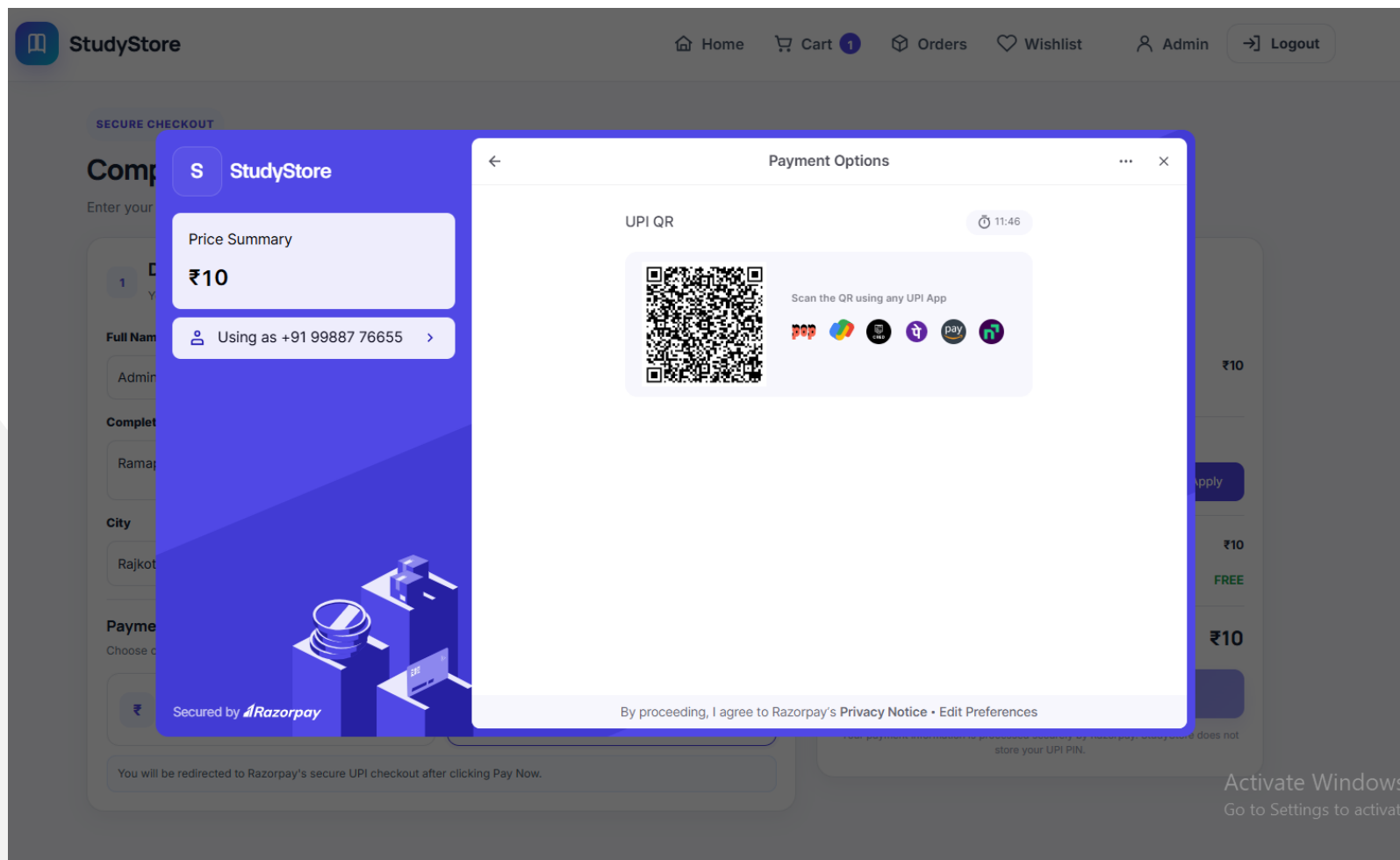
Pay ₹10 with UPI

Your payment information is processed securely by Razorpay. StudyStore does not store your UPI PIN.

Activate Window
Go to Settings to activate

Screen Design

Razorpay UPI Payment Screen





Project Overview

1. User Registration & Login

- Users can create a new account with their details.
- JWT authentication protects user sessions; passwords are hashed with bcryptjs.
- Users can manage their profile information.

2. Product Browsing, Search & Wishlist

- Customers browse products with category filters and keyword search.
- Products can be saved to a personal wishlist for future purchase.

3. Shopping Cart & Checkout

- Products can be added to the cart, with quantities updated before checkout.
 - Checkout supports Cash on Delivery and secure UPI payments via Razorpay.
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Project Overview

4. Order Placement & Tracking

- Orders are created from the shopping cart and stored in MongoDB Atlas.
- Customers can view order history and track current order status.

5. Admin Dashboard

- Administrators can add, edit, or delete products and manage stock.
- Order status can be updated, and discount coupons can be created and managed.

6. Password Recovery

- Users can reset a forgotten password through an email link sent using Nodemailer.

System Testing

Test Cases (TC-01 to TC-05)

Test ID	Module	Scenario	Expected Result	Status
TC-01	Registration	Register with valid details	Account created successfully	Pass
TC-02	Login	Login with valid credentials	User logged in successfully	Pass
TC-03	Forgot Password	Enter registered email	Reset link sent to email	Pass
TC-04	Product Search	Search product by keyword	Matching products displayed	Pass
TC-05	Wishlist	Add product to wishlist	Product added to wishlist	Pass

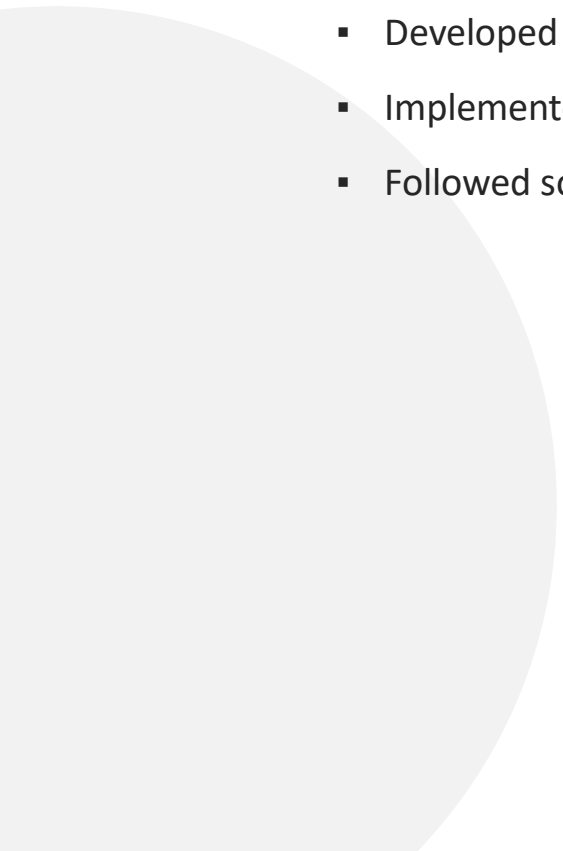
System Testing

Test Cases (TC-06 to TC-10)

Test ID	Module	Scenario	Expected Result	Status
TC-06	Shopping Cart	Add product to cart	Product added to cart successfully	Pass
TC-07	Place Order	Place order from cart	Order placed and stored successfully	Pass
TC-08	My Orders	View previously placed orders	Order history displayed correctly	Pass
TC-09	Product Management	Admin adds/edits/deletes product	Product updated successfully	Pass
TC-10	Order Management	Admin updates order status	Updated status displayed correctly	Pass



Conclusion

- Successfully developed a complete online stationery shopping application using the MERN Stack.
 - Implemented customer features: registration, login, product browsing, wishlist, cart, and order placement.
 - Designed an administrator dashboard for efficient product and order management.
 - Applied JWT authentication, bcryptjs password encryption, and Nodemailer for account management.
 - Developed a structured MongoDB Atlas database for efficient data storage and retrieval.
 - Implemented RESTful APIs enabling smooth communication between frontend and backend.
 - Followed software engineering practices including system design, UML diagrams, and testing.
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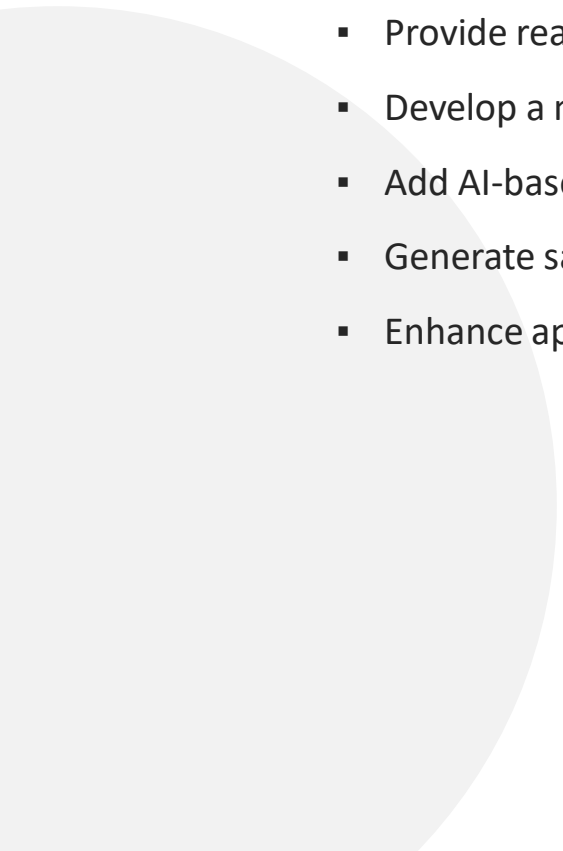


Learning During Project Work

- Gained practical experience developing a full-stack web application using the MERN Stack.
 - Learned to design responsive user interfaces using React.js and Vite.
 - Improved backend development skills using Node.js and Express.js.
 - Acquired hands-on experience with MongoDB Atlas for cloud database management.
 - Learned JWT-based authentication, password encryption, and email-based password recovery.
 - Improved knowledge of Git and GitHub for version control and collaboration.
 - Enhanced understanding of UML diagrams, database design, and software testing.
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Future Enhancement

- Integrate additional online payment gateways such as Stripe.
 - Implement inventory and stock management with low-stock notifications.
 - Add product ratings and customer reviews.
 - Introduce order cancellation and return request features.
 - Provide real-time order tracking and email notifications.
 - Develop a mobile application using React Native.
 - Add AI-based product recommendations based on user preferences.
 - Generate sales reports and analytics for administrators.
 - Enhance application security with Two-Factor Authentication (2FA).
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Thank You

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Chirag Pandya - 92500584070



Thank
You

